



Subject Title: Strength of Materials

Course Title: B.Eng. Manufacturing Engineering

Report Title: Hardness Testing

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Declaration: I hereby declare that the work outlined above has been completed by myself, it has not been shared with any other student nor has it been copied from other sources.

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Introduction

Hardness is a material ability to withstand friction, essentially abrasion resistance. hardness testing is commonly used to evaluate some of the key mechanical properties of materials. For example, the hardness value of a metallic material enables the estimation of its wear resistance, ductility, flow and cut tension. Hardness is a measure of how easily a material can be scratched or indented. Hard materials are often also very brittle. This means they have a low resistance to impact and easily scatter. Hardness testing is widely used for material evaluation because of its simplicity and low-cost relative to direct measurement of many properties. Specifically, conversion charts from Rockwell hardness to tensile strength are available for some structural alloys, including steel and aluminium. Based on the idea that a material's response to a load placed at one small point is related to its ability to deform permanently, the hardness test is performed by pressing a hardened steel ball (Brinell test) or a steel or diamond cone (Rockwell test) into the surface of the test piece. Most hardness tests are performed on commercial machines that register arbitrary values in inverse relation to the depth of penetration of the ball or cone.

Aims/Objectives

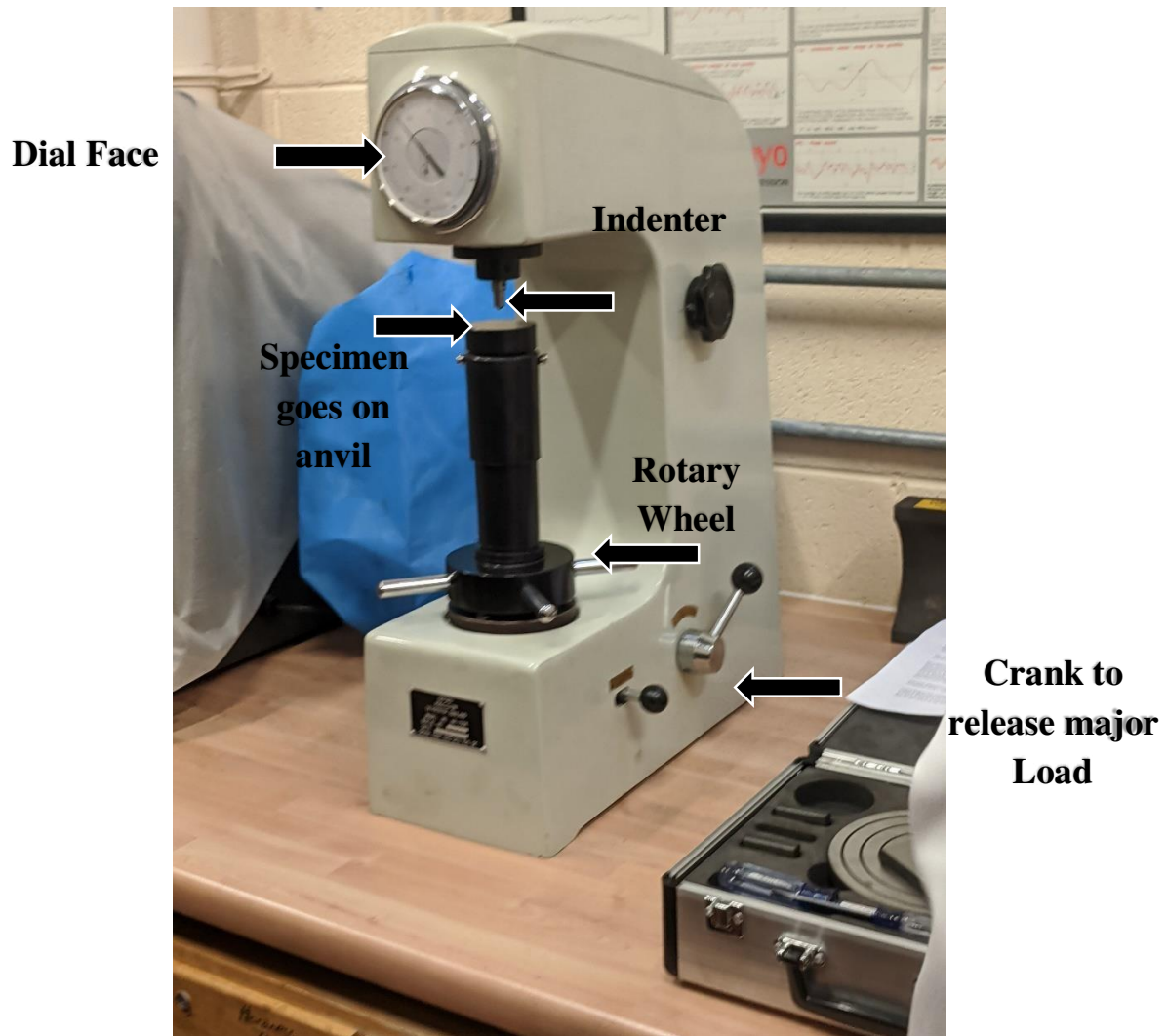
Aims:

- To measure the hardness of various specimens using the Rockwell method and Brinell method.
- Being able to measure the size of the indent on each of the specimens.
- To record the results during the experiment of each test we conduct for later calculations.

Objectives:

- Write a report on Hardness testing using two methods Rockwell and Brinell and describe them using different headings including procedure, Apparatus, results and come up with a conclusion to discuss possible errors and improvements.
- Comment on the combined results that how specimens faired on the impact test and then how they performed under the Rockwell hardness test).
- Compare the hardness numbers of the Z7 specimens at different loads.

Apparatus



Rockwell Hardness Tester

Procedure For Rockwell Hardness Testing

- Firstly, the specimen was put on anvil in contact with the indenter.
- A minor load of 10 kg was applied and the dial is set to zero and then major load is applied.
- Preliminary force (10kg for normal and 3kg for superficial testing) to a penetration depth of h_0 in the specimen to be tested.
- H_0 refers to level for subsequent measurement of the residual indentation depth(h).
- Automatically or manually zeroing the penetration measuring instrument or dial indicator.
- After that, additional test force (60,100 or 150 kg for normal and 15, 30, 45kg for superficial testing) is applied for a dwell period (15seconds).
- Machine will show the Rockwell Hardness Number HR on the machine.
- Later on, gently remove the additional force while leaving in place with the preliminary force so that the elastic deformation follow the removal of final force is recovered.
- After that the depth of penetration was measured using the instrument.
- Furthermore, Rockwell hardness can be calculated using the residual indentation depth (h) and a formula defined in the standard, taking account of the applied Rockwell scale.
- Rockwell hardness testing is used to test hardness in the macro range test force within 49.03 newton (5kg) to 1471 newton (150kg).
- It is a differential-depth method which used the residual depth of indentation left by the indenter is measure to determine the hardness value of a test specimen.
- Indenter can be either diamond cone (cone angle of 120°) or a hard metal ball (with various diameter depending on the method 1/16", 1/8" being common).

Results

Rockwell		
Scale C	Scale B	Scale A
150kgf	100kgf	60kgf

Hardness Testing	
Rockwell Testing	
Scale C set to 150Kg	
Specimen used in test	Size of indent
Z-specimen	C10
Z4-specimen	C34

Hardness Testing	
Rockwell Testing	
Scale B set to 100Kg	
Specimen used in test	Size of indent
Z-specimen	B95.5

Source of errors

- Impact of pressure - strength and hardness carbide ball is not enough, which is prone to deform. Carbide ball is flattened and deform into oval permanently.
- Placement of the hardness is not correct

- Surface of the specimen contact badly with the table or support point is not stable, which produce slippage, rolling, tilt and etc.
- Environment- vibration can impact the loading
- Human error - experience of the operator is not mature enough. Exert test force is too fast, holding time is short and the load is too slow to maintain a long time.

Apparatus

Brinell Hardness Testing



Universal Tester



Load Meter

Procedure For Brinell Hardness Testing

- Brinell hardness testing is typically done on iron and steel casting using a 3000kg test force and a 10mm diameter carbide ball. While softer material used 500kg test force and 10 or 5mm carbide ball.
- Firstly, indenter is pressed into the sample by an accurately controlled test force.
- The force is maintained for a specific dwell time (15seconds).
- After the dwell time is complete, the indenter is removed leaving a round indent in the specimen.
- The size of the indent is determined optically by measuring two diagonals of the round indent using either a portable optical microscope or one that is integrated with the load application device.
- The Brinell hardness number is a function of the test force divided by the curved surface area of the indent.
- The indentation is considered to be spherical with a radius equal to half the diameter of the ball.

Results

Hardness Testing	
Brinell Testing	
Set to 29.5KN	
Specimen used in test	Size of indent (mm)Ø
Z-specimen	4.6
Z4-specimen	2.64
Z7-Specimen at 3000kg	4.3
Z7-Specimen at 1500kg	3.3

Source of error

- Human error- operator not enough experience to operate accurately.
- Environment- vibration can cause loading accuracy, lighting which we need to use portable optical microscope to measure the indentation.
- Surface of the indentation contact badly with the machine.
- Dwell timing not enough.
- Sample preparation- specimen should be test surface free of scratches and surface texture.

Calculations

Calculations of the Brinell hardness value is carried out by using the following formula:

$$HBW = 0.102 \frac{2F}{\pi D (D - \sqrt{D^2 - d^2})}$$

where:

F = applied load (Newtons)

D = diameter of indenter (mm)

d = diameter of indentation (mm)

Z Specimen

29.5KN to Newtons = $29.5 \times 1000 = 29500\text{N}$

Here

is the specimen Z with a force of 29.5 KN applied.

$$HB = \frac{(0.102) \frac{2(29500)}{\pi(10) (10 - \sqrt{(10)^2 - (4.6)^2})}}{}$$

HB value = 170.91

Z4 Specimen

29.5KN to Newtons = $29.5 \times 1000 = 29500\text{N}$

$$HB = \frac{(0.102) \frac{2(29500)}{\pi(10) (10 - \sqrt{(10)^2 - (2.64)^2})}}{}$$

HB value = 541.22

Z7 Specimen

So, HB for specimen Z7 with 3000kg force is $3000\text{kg} \times 9.81 = 29430\text{ N}$

$$HB \approx 0.102 \times \frac{2(29430\text{N})}{\pi(10)(10 - \sqrt{10^2 - 4.2^2})} \quad \text{HB value} = 206.87$$

So, HB for Specimen Z7 with 1500kg force is $1500\text{kg} \times 9.81 = 14715$

$$HB \approx 0.102 \times \frac{2(14715\text{N})}{\pi(10)(10 - \sqrt{10^2 - 3.3^2})} \quad \text{HB value} = 170.35$$

Comparing

Specimen	Size of indent (mm)Ø	Load
<i>Z7</i>	4.3	3000kg
<i>Z7</i>	3.3	1500kg

Z7 specimen at a load of 3000kg have a indent of 4.3mm and the same specimen with a different load of 1500kg have a indent of 3.3mm so the difference between them is 1mm the main reason behind this difference is the different loads used for each of them, another reason can be as discussed above is the Surface of the indentation contact badly with the machine or the dwell timing was not enough.

Conclusion

This experiment was a complete success as accurate enough results were obtained although results were accurate. It can be seen from readings which are obtained in this experiment using two techniques rockwell and brinell hardness testing that the expected values and the obtained values were very much similar and there were very small difference which could be attributed to experimental errors as mentioned above such as the human error in reading the values or Environment- vibration which can cause error in the results. To myself after taking part in this experiment I learned a lot about hardness testing, with the help of rockwell and brinell hardness test, that how we can measure the hardness of each different type of material and how both type of method measured the hardness but using different kind tools.

References

_Notes up on Black Board [Accessed 1 May 2020]

_ <https://www.mee-inc.com/hamm/rockwell-hardness-testing/> [Accessed 2 May 2020]